

Challenges in Interpretation of US Breast Findings in the Emergency Setting

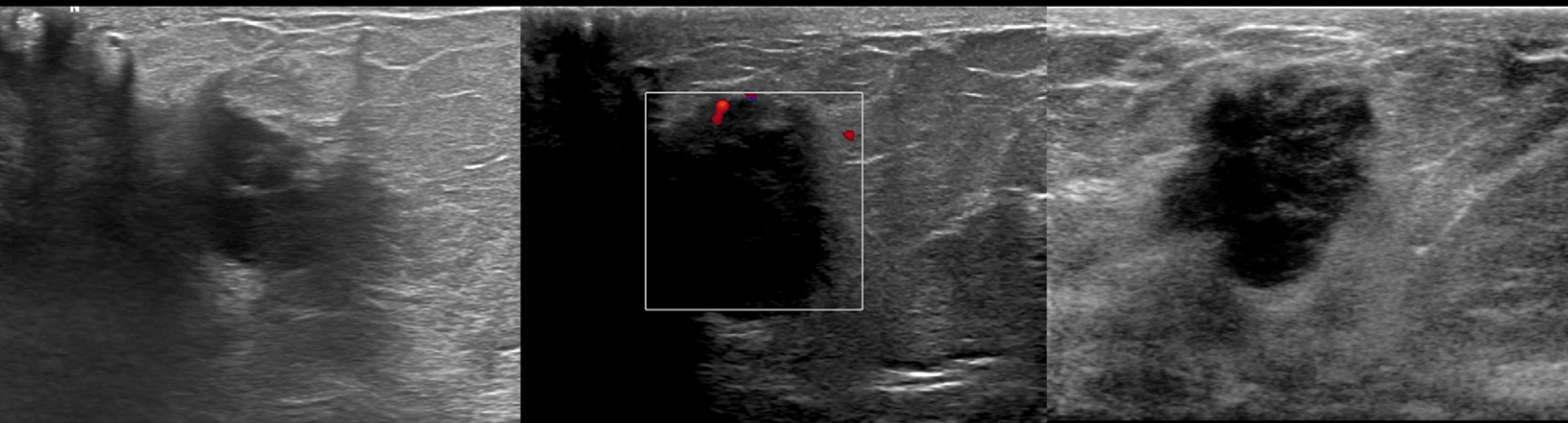
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Emergencies in breast imaging are infrequent but not rare. Although infectious conditions such as mastitis and breast abscess are the most common breast diseases encountered in acute care settings, other entities that may require additional imaging or different treatment approaches are also seen and include traumatic injury and breast cancer. While mammography is widely available for breast evaluation in outpatient facilities, most emergency departments do not have mammography units. This makes evaluation of patients with breast disease incomplete in the acute care setting and emphasizes the role of appropriate US techniques for interpretation. It also highlights the importance of effective sonographer-to-radiologist communication to ensure patient safety and diagnostic accuracy, especially in an era of increasing adoption of teleradiology. The authors discuss the challenges in image acquisition and remote interpretation that are commonly faced by radiologists when they assess breast anomalies in the emergency setting. They present strategies to overcome these challenges by describing techniques for proper US evaluation, highlighting the importance of sonographer-radiologist communication, defining the goals of the evaluation, reviewing common differential diagnoses, and providing appropriate follow-up recommendations.

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Introduction

Evaluation of patients with breast symptoms in the emergency department (ED) can be a challenge for ED radiologists, especially those without specialized training in breast imaging. Infectious conditions such as mastitis and breast abscesses are the most common breast diseases seen in the acute care setting. However, other entities such as traumatic injury, postsurgical complications, and physiologic conditions may also be encoun-

tered. Some of these conditions may have clinical manifestations that overlap with those of breast cancer, highlighting the importance of accurate characterization of imaging features and appropriate follow-up recommendations.

The goals of radiologists in the ED are to identify the cause of the patient's breast-related symptoms, recognize entities that require urgent treatment in the emergency setting, give appropriate recommendations to ensure clinical and imaging

Supplemental Material



Quiz questions for this article are available in the supplemental material.

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Content Codes: BR, ER

Abbreviations: BI-RADS = Breast Imaging Reporting and Data System, ED = emergency department, IBC = inflammatory breast cancer

TEACHING POINTS

- Clear and consistent bidirectional communication between the radiologist and sonographer is necessary and must be achievable around the clock.
- Just as important is the role of the ED radiologist to recognize that these commonly seen entities may have symptoms that overlap with those of breast cancer, and if the latter is a possibility, then immediate specialized assessment is necessary and must be recommended.
- Owing to the significant overlap in the clinical and imaging findings of mastitis and inflammatory breast cancer (IBC), the most lethal form of breast cancer, close monitoring of symptomatic response to antibiotic therapy in patients with mastitis is strongly recommended.
- At US, abscesses may have posterior acoustic enhancement, a thick irregular echogenic rim, and increased peripheral vascularity, hampering their differentiation from necrotic or cystic tumors. Studies have demonstrated an association of posterior acoustic enhancement with high-grade and hormone receptor-negative breast tumors, causing additional overlapping of imaging features between these neoplasms and breast abscesses.
- Patients of any age who have clinical findings that are concerning for malignancy but negative US findings must be promptly referred to the breast imaging department for complete imaging evaluation and possible biopsy.

resolution of symptoms, and provide a prompt evaluation in the breast imaging department, as necessary, to exclude the possibility of malignancy.

While mammography is widely available for breast evaluation, most EDs do not have mammography units. This issue emphasizes the role of appropriate US techniques and the importance of effective sonographer-to-radiologist communication to ensure appropriate patient management. The latter is especially important because radiologists are usually unable to perform real-time imaging of patients in this setting. Many additional challenges and limitations, ranging from inadequate access to patients' medical and clinical histories to suboptimal follow-up and navigation systems, further impede optimal management and timely follow-up and are discussed in this article.

Challenges in Evaluation in the Emergency Setting

Challenges in the evaluation of breast findings in the ED are multifactorial. These variables can be categorized in terms of equipment availability, technologist and imaging technique, technology and telemedicine limitations, clinical and medical history limitations, limited Breast Imaging Reporting and Data System (BI-RADS) utilization, and lack of follow-up and patient navigation systems.

Equipment Availability

US is a noninvasive, nonionizing, and inexpensive imaging modality that is well tolerated by patients with acute breast disease and is readily available in the emergency setting. It has high sensitivity for the diagnosis of fluid collections,

which are commonly encountered in the ED (1). However, compared with US, mammography has higher sensitivity and specificity in the evaluation of suspicious masses and calcifications, especially in patients without dense breast tissue (2–4). In women younger than 30 years who present with palpable abnormalities or focal pain, US alone has been shown to have a negative predictive value of 100% in examinations performed by breast imaging radiologists. Although this result cannot be extrapolated to the emergency setting, it highlights the advantage that US provides in ruling out malignancy when it is performed as the primary imaging examination in this age group (5).

According to current American College of Radiology appropriateness criteria recommendations, for symptomatic patients aged 40 years or older, mammography is the most appropriate initial imaging modality. In addition, in patients aged 30–39 years, mammography can be the initial or complementary examination performed to evaluate for breast symptoms. The lack of mammography in the emergency setting results in incomplete workups for patients within these age ranges who present to the ED (6,7).

Technologists and Imaging Technique

Unlike in most breast imaging centers, where US is performed by dedicated breast imaging-trained sonographers with direct supervision by fellowship-trained breast imaging radiologists, US in the ED is commonly performed by general sonographers with remote radiologist supervision (8–12). The lack of subspecialty expertise may be associated with misdiagnoses and technically suboptimal studies (12,13). In addition, a sonographer's limited experience in breast US may result in inadequate image quality due to improper and incomplete image acquisition, incorrect or incomplete image labeling, and inappropriate patient positioning, among other causes.

Technology and Telemedicine Limitations

During the past 2 decades, teleradiology has evolved to a point where 85.6% of radiologists in the United States report being engaged in this practice (14). Interpretation of radiologic studies from the ED is one of the most popular services that radiologists provide through teleradiology (14,15). Lack of technologist proximity has been identified as a major challenge by up to 48.4% of teleradiologists (14). In the emergency setting, in which study interpretations performed off-site and imaging areas isolated from the reading rooms are common practice, the US technologist serves as an irreplaceable link between the medical equipment, patient, and radiologist (16). The synchronous image acquisition model, which enables real-time communication between the technologist and the radiologist, results in better image quality control and a reduced need for additional images. However, use of this model, as compared with the asynchronous (ie, store-and-forward) model, in the emergency setting is technically and logistically challenging (17,18).

Limited communication with referring providers also represents a major limitation of teleradiology (14,19,20). Verbal communication facilitates the exchange of complex information and appropriate patient prioritization and is more likely to have a positive impact on clinical decision making (19–22).

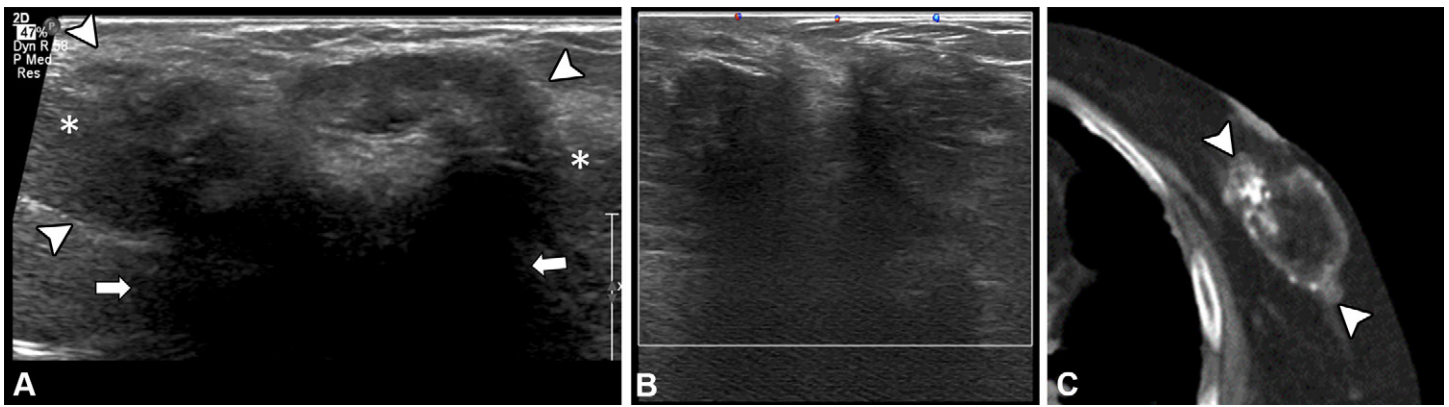


Figure 1. Left breast lump in a 61-year-old woman who presented to the ED. (A, B) Gray-scale (A) and color Doppler (B) US images obtained in the ED show an oval heterogeneous left breast mass (arrowheads in A) that has indistinct margins (* in A) and posterior acoustic shadowing (arrows in A), correlating with a palpable area of concern, with no internal vascularity seen in B. (C) Recent abdominal CT image prior to presentation shows a partially visualized, large, fat-containing mass (arrowheads) with dystrophic calcifications in the left breast, consistent with fat necrosis. The patient confirmed having a previous history of reduction mammoplasty. This case highlights the importance of the availability of previous imaging studies for comparison.

Clinical and Medical History Limitations

The quality and clinical relevance of the radiologic evaluation for breast disease depend heavily on the patient's clinical presentation, medical information, and current versus prior imaging findings, as well as the radiologist's understanding of the patient's clinical concerns (10). It is not uncommon for radiologists to receive imaging requisitions that have incomplete clinical information and lack pertinent physical examination findings that are necessary for appropriate image interpretation. These missing data reduce the accuracy of the radiologic report and subsequently affect patient care (23,24) (Fig 1).

Limited Utilization of BI-RADS

The use of a standardized reporting system allows better outcome monitoring; prevents confusion in interpretation; and facilitates communication with clinicians, radiologists, and patients (25–27). These are the purposes of BI-RADS, which is widely used to describe breast imaging findings. However, the use of BI-RADS terms in examinations performed in the ED is not a routine practice, and, thus, those examinations cannot be properly audited (9,25).

Lack of Follow-up and Patient Navigation Systems

Although the American College of Emergency Physicians supports the use of patient navigators in the ED (28), the interventions aimed at accelerating the transition of patients from the ED to outpatient care vary broadly across the country and different health care systems. The most described ED-based initiatives are written instructions given to the patient at discharge, scheduling a follow-up appointment before discharge, telephone contact or automated text message appointment reminders, alerting outpatient providers, and more complex care coordination services such as patient navigation (29).

Patient navigation is a patient-centered approach used by clinical professionals and nonclinical health care workers who are integrated into the health care team to improve the continuity of patient care by identifying and addressing indi-

vidual barriers to care (30,31). Patient navigation is a common practice in breast imaging (31). It has been shown to be effective in increasing the screening rate and decreasing the times to diagnostic follow-up and treatment initiation (31–33). For patients with breast findings, the lack of patient navigators and patient reminder systems in the ED can result in loss of follow-up and delays in care, which negatively affect patient outcomes (33).

Overcoming the Challenges

Several strategies can be implemented to overcome the described challenges and include equipment and technical improvements, technologist training, and system-wide initiatives (Table 1).

Equipment

US equipment must include a matrix or linear transducer with a wide footprint, and it is recommended that it have a center frequency of at least 12 MHz, which is suitable for evaluation of most breasts (10,34,35). Transducers with a lower frequency and further penetration capability are necessary when evaluating large breasts (>5 cm in thickness) in an appropriately positioned patient (34).

Stand-off acoustic pads play a significant role in the evaluation of superficial lesions by bringing them closer to the transducer focal zone (10). Foam wedges are helpful for trunk stabilization while maintaining optimal patient positioning during the US examination. In addition, equipping the US room with a dimmer switch allows the lighting adjustments necessary to decrease ambient light reflections and increase the contrast sensitivity of the sonographer's eyes.

Technical Improvements

Imaging Technique.—Gain, depth, focal zone, and time-gain compensation can be manually tailored to optimize images; this capability highlights the importance of the technologist's basic training and knowledge of this anatomic region. Initial gray-scale gain settings should be adjusted so that fat lobules

Table 1: Interventions to Improve Breast US Interpretation in the ED

Challenge	Interventions
Equipment availability	Linear transducer with a frequency of at least 12 MHz Warm gel Wedge and/or pillow for support Stand-off acoustic pad Appropriate room lighting
Technologist	Undergo basic training to increase familiarity with the anatomic region Use clear image annotations Image findings in two orthogonal planes and use color Doppler imaging Complete ED US technologist form (Fig 7) Continuous case review and education performed by breast imaging department staff
Technique	Apply optimal gain and depth settings Position focal zone at level of or just below finding Use tissue harmonics and spatial compounding techniques Create a custom preset for breast evaluation Position patient appropriately
Radiologist	Recognize common differential diagnoses in the ED Provide appropriate follow-up recommendations Be available to immediately address sonographer's questions Provide feedback to and perform case review with the technologist
System	Refer suspicious cases to the breast imaging department navigation group Schedule follow-up appointments in the breast imaging department before patient's discharge from the ED Promptly notify the breast radiologist if urgent intervention is necessary

appear medium gray (34). If the gain is set inappropriately high, the appearance of extensive internal echoes may significantly limit the ability to differentiate a simple cyst from complex cystic and solid lesions.

On the contrary, if the gain is set abnormally low, a solid mass may be mistaken for a cyst (35). The time-gain compensation should be adjusted to equalize the echogenicity of tissues at all depths (34). While the depth is being adjusted, the pectoralis muscle must be visualized along the posterior margin of the image to ensure the inclusion of all three zones—premammary, mammary, and retromammary—of the breast (34,36). Positioning the focal zone at the same level of or just below the area of interest helps to optimize the resolution and display posterior acoustic features of the lesion (35,36). A custom preset with settings optimal for breast tissue may be created and used as a reference point, especially by less experienced sonographers who are exposed to the fast-paced environment of the ED.

Spatial compound imaging increases image quality owing to the reduction of speckle noise and posterior acoustic artifacts, allowing better identification of microcalcifications and better characterization of lesion borders (34,37). Tissue harmonics substantially improve clutter noise, eliminate artifactual internal echoes from fluid-filled lesions, facilitate definition of the lesion border, and enhance acoustic shadow conspicuity (34,37). Both spatial compound imaging and tissue harmonics should be routinely used for characterization of encountered abnormalities (Fig 2). The speed of sound may be adjusted, depending on the breast density, as speed propagation varies according to the amount of fat or glandu-

lar tissue in the breast. This enables better depiction of calcifications and better characterization of lesion margins (34).

When differentiating solid from cystic lesions, gain and focal zone optimization, tissue harmonics, spatial compounding, and color Doppler imaging are beneficial (34–37). Spatial compounding, tissue harmonics, and sound speed adjustments may be applied for better margin delineation (Fig 3) (34,37).

Positioning and Patient Variables.—Inadequate patient positioning may create unnecessary artifacts, obscure small lesions, and affect the reproducibility of lesion localization or measurements. The goal of optimal positioning is to minimize the breast thickness in the area being imaged (10). Patients are encouraged to lie supine with the ipsilateral arm over the head. For more lateral lesions, a contralateral supine oblique position, which can be achieved with wedge pillow support, is recommended (34). The lateral decubitus position may be necessary for patients with large breasts. Since small vessels within the breast are easily compressible, an appropriate amount of pressure must be applied while scanning to prevent underestimation of blood flow on color Doppler US images (35).

A satisfactory amount of gel is necessary to ensure a lack of air between the skin and the probe. Air in the space between the skin and the probe results in artifacts and limited visualization of structures (Fig 4) (34,36). The application of warm gel not only enhances patient comfort but also plays a role in optimizing breast tissue evaluation, especially the subareolar region, reducing nipple muscular contraction and facilitating ducts and glandular tissue expansion (Fig 5) (38).

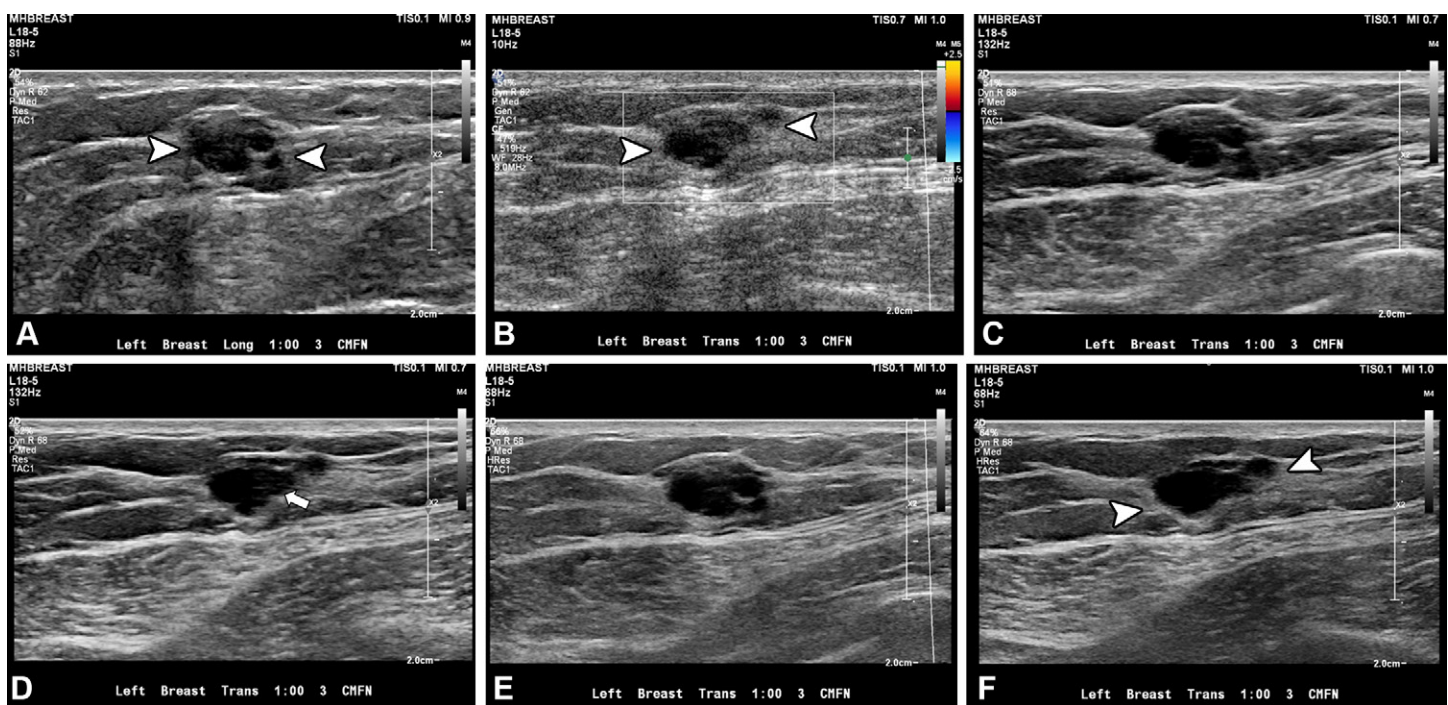


Figure 2. Palpable painful mass in the left breast of a 40-year-old woman who presented to the ED. (A, B) Initial longitudinal (A) and transverse color Doppler (B) US images show a hypoechoic, lobulated, complicated cyst or complex mass (arrowheads) without increased vascularity in the left breast at the 1-o'clock position. (C, D) Transverse US evaluation with spatial compounding resulted in a significant reduction in speckle noise in the breast parenchyma and better delineation of the lesion margins, with the artifactual internal echoes (arrow in D) still visible. (E, F) Further transverse US evaluation with tissue harmonics revealed a reduction in artifactual internal echoes (arrowheads in F) and better margin delineation, favoring a cluster of complicated cysts. After further questioning, the patient reported previous trauma to this region, and fat necrosis was included in the differential diagnosis.

Technologists.—US technologists must have familiarity with the standardized protocol for breast US techniques, patient positioning, and image labeling. To ensure the reproducibility of lesion localization, technologists must add not only the basic annotations such as laterality and plane (eg, transverse or longitudinal, radial or antiradial) on images but also the clock-face position of the lesion and the distance from the nipple in centimeters. The location of the lesion may also be reported by using a clock-face diagram that can be submitted to the picture archiving and communication system. Lesions should be measured in three dimensions in two orthogonal planes, and images with and without calipers must be recorded, as images without calipers allow complete margin assessment (10). Color Doppler US images of lesions must be obtained for evaluation of internal vascularity (Fig 6) (35,36).

Radiologist and Sonographer Communication.—The US technologist's communication of important clinical information and imaging findings is a crucial component in the evaluation of breast-related emergencies. Clear and consistent bidirectional communication between the radiologist and sonographer is necessary and must be achievable around the clock. To ensure that this communication is faultless, we suggest the use of a sonographer-standardized form that includes spaces assigned for patient concerns, pertinent medical history, important physical observations, study limitations, and US finding annotations on a clock-face breast diagram (Fig 7).

Real-time scanning is an advantage when evaluating the breast with US, as it permits evaluation of lesion mobility and subtle findings that may not be detectable on static images

and aids in differentiating floating debris within fluid collections from fixed solid components within complex lesions (34,39). In addition, dynamic color Doppler US may depict internal vascularity or feeding vessels that are difficult to capture on static images and thus raise the level of suspicion for malignancy (39). The lack of this capability for radiologists in the ED is a limitation. However, the use of US cine loops in these situations may be beneficial, as investigators in previous studies (40,41) have reported faster examination performance by sonographers, more accurate recognition of artifacts by interpreting radiologists, and reduced need for repeat imaging.

Decreased face-to-face communication between radiologists and technologists reduces opportunities for interpersonal communication, timely feedback, and education, impacting patient care (20). By reviewing the technologist's work and providing feedback, radiologists can help technologists improve their US skills. This feedback can be implemented regularly with scheduled meetings, case conferences, and digital forms of communication. Involving lead technologists in the design of the ED breast US protocol, implementing the protocol throughout the institution, and implementing quality control processes ensure effective communication and accurate image acquisition (42).

Systems for Evaluation and Follow-up.—Efforts to overcome the technical challenges in the ED must be complemented by an effective system of notification and patient navigation services. Tools for closed-loop communication within the picture archiving and communication system and the electronic health record may be included in the ED workflow to notify patients

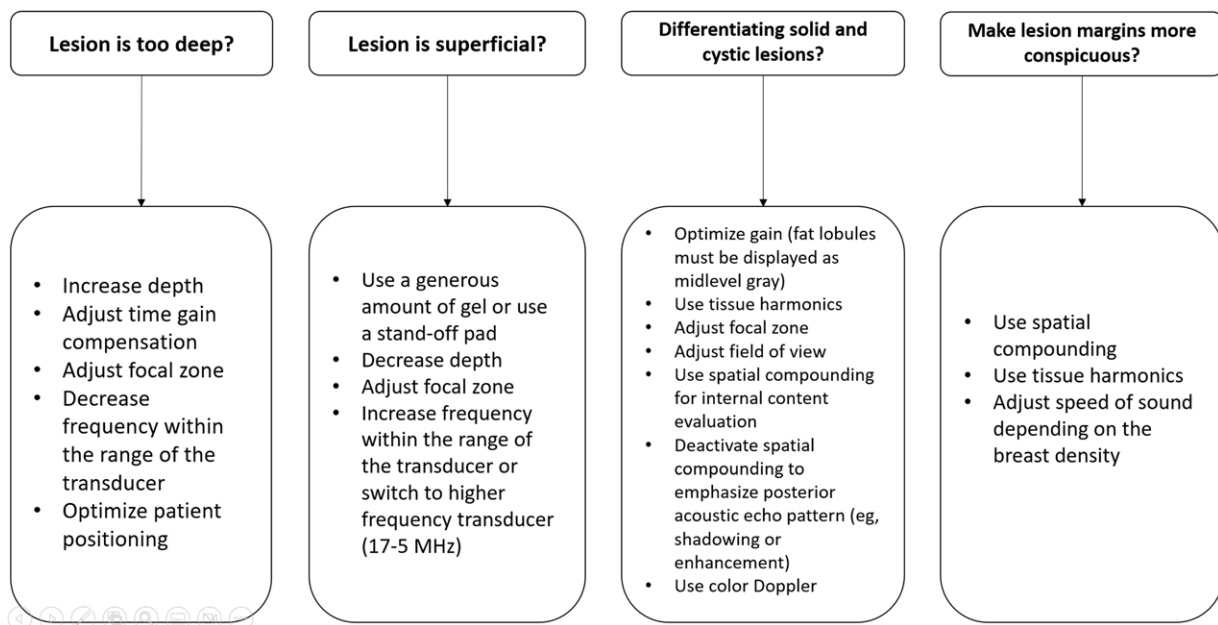


Figure 3. Troubleshooting algorithm at breast US image acquisition.

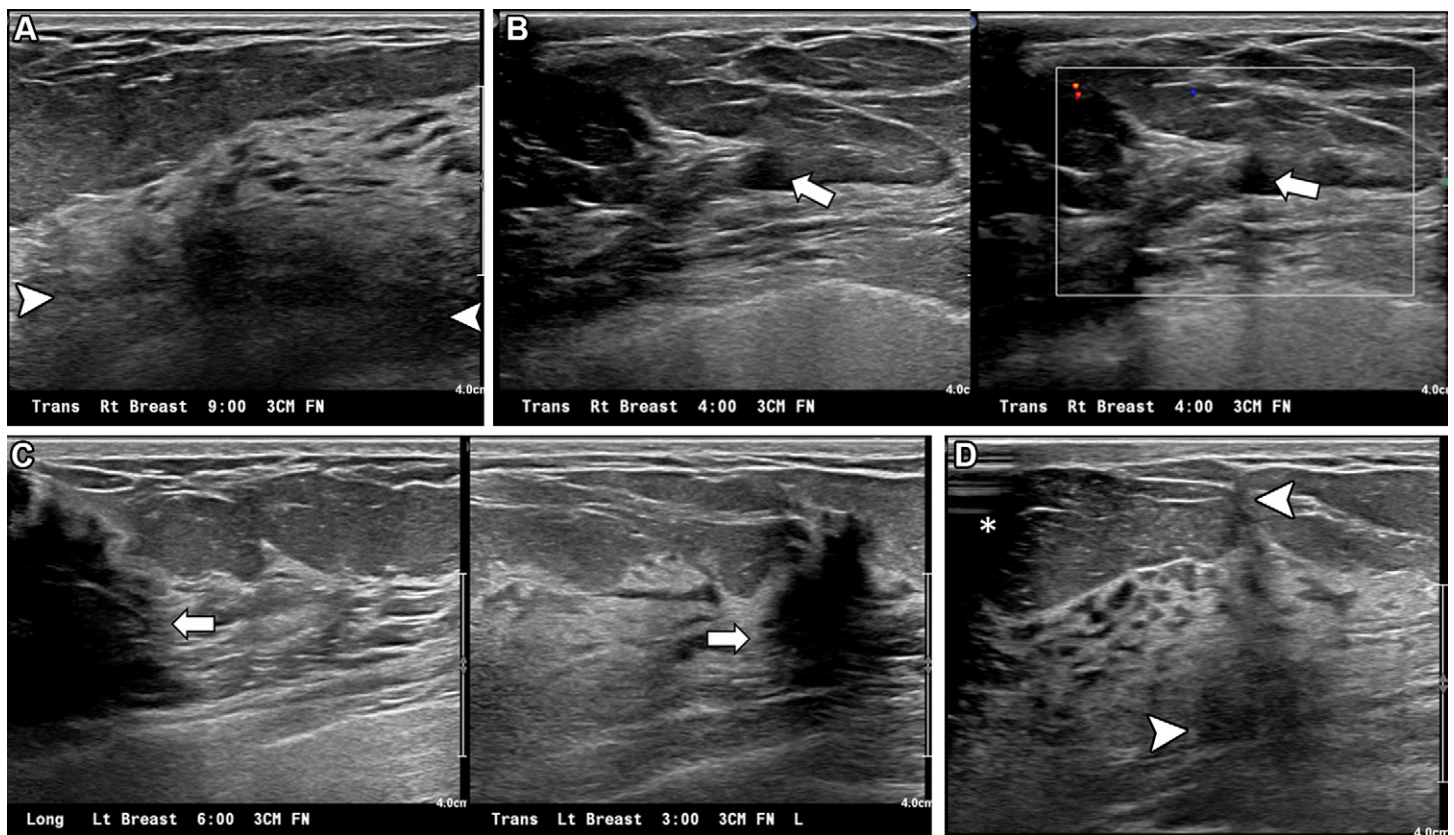


Figure 4. Bilateral breast pain, which was greater on the right than on the left, in a 51-year-old woman who presented to the ED. (A, B) Transverse US images of the right breast show inappropriate gain and depth that limited the evaluation of deep breast tissue (arrowheads in A), as well as shadowing and pseudomasses (arrows in B) caused by a lack of pressure. (C, D) Similar changes were noted on longitudinal (left image in C) and transverse (right image in C; D) US images of the contralateral breast and included pseudomasses (arrows in C), shadowing from Cooper ligaments (arrowheads in D), and insufficient gel (*) in D). Repeat US performed in the breast imaging department (not shown) showed negative and unremarkable findings.

and referring physicians of follow-up recommendations and improve follow-up completion rates (43). For studies that are performed after hours and require immediate or urgent man-

agement by the breast imaging department, a system of notifying the breast imaging team must be implemented to ensure rapid evaluation. In addition, patients with breast symptoms

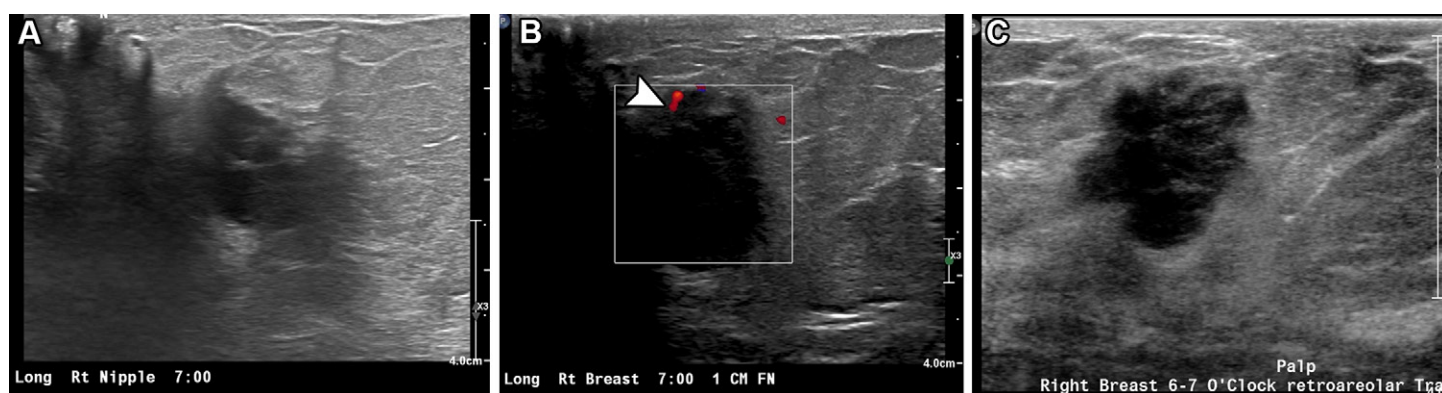


Figure 5. Palpable mass and pain for 1 week in the right breast of a 51-year-old woman who presented to the ED. (A) Longitudinal US image obtained in the ED shows a large area of heterogeneity along the 7- to 9-o'clock axis in the right breast, with posterior acoustic shadowing. (B) Longitudinal color Doppler US image of the subareolar region shows vascularity (arrowhead) within a more distinct area of hypoechogenicity. The findings, although inconclusive, were suspicious, and further evaluation in the breast imaging department was recommended. (C) Transverse US image obtained in the breast imaging department shows an irregular antiparallel mass in the right breast in the 7-o'clock position, 1 cm from the nipple, with increased vascularity. The previously noted large area of heterogeneity in the right breast seen on the ED image (A) was not present at breast imaging department US evaluation and was secondary to suboptimal imaging technique, with shadowing caused by a lack of appropriate pressure and incorrect positioning of the patient, incorrect depth and gain, and shadowing from the nipple due to a lack of gel and nipple contraction. Biopsy of the mass revealed invasive ductal carcinoma.

Quick initial breast US image quality assessment

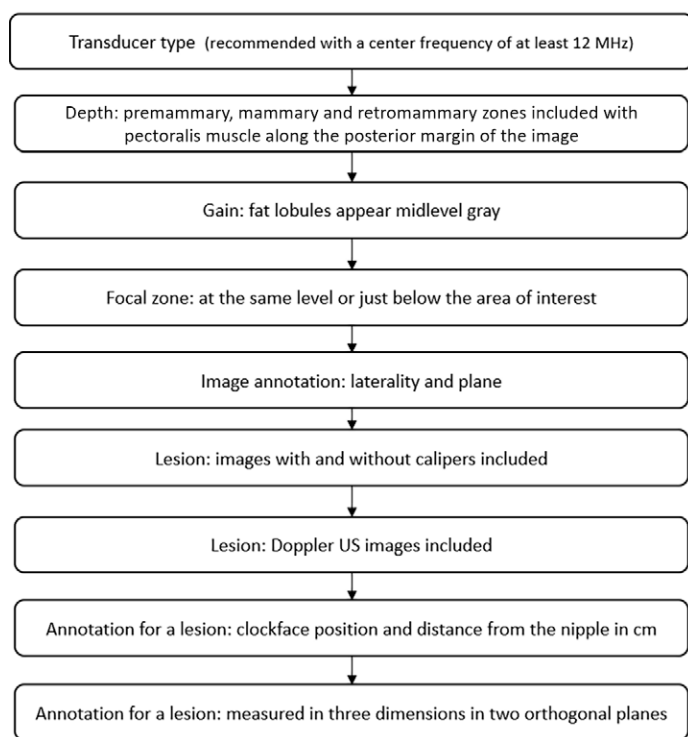


Figure 6. Steps for quick assessment of initial breast US image quality.

and follow-up recommendations must receive contact information (name, location, phone number) for the breast imaging department and potentially have a follow-up appointment scheduled before they are discharged from the ED.

We suggest including suspicious cases in the breast imaging department's internal navigation group referrals to facilitate patient adherence to follow-up and timely access to optimal patient care (31,33). This is especially important for patients whose social determinants of health (eg, living in a

socioeconomically disadvantaged neighborhood, lack of insurance, and low health literacy) have a negative impact on follow-up compliance (29,43).

Goals of Evaluation and Common Differential Diagnoses in the ED

Although breast-related emergencies are uncommon, radiologists in the ED must be familiar with commonly encountered breast diseases and the associated sonographic findings and be able to identify a cause of the patient's breast-related symptoms, recognize entities that require immediate management, and provide appropriate follow-up recommendations. Just as important is the role of the ED radiologist to recognize that these commonly seen entities may have symptoms that overlap with those of breast cancer, and if the latter is a possibility, then immediate specialized assessment is necessary and must be recommended. Breast-related emergencies can be divided into those that require immediate treatment and those that are nonurgent but require prompt specialty evaluation or patient reassurance.

Breast-related Emergencies Requiring Immediate Treatment

Emergent breast conditions that require immediate treatment usually fall into one of three categories that are based on the cause of the entity: infectious and/or inflammatory, postsurgical or postprocedural, or trauma (Fig 8).

Infectious and/or Inflammatory Conditions

Infection and inflammation of the breast parenchyma are commonly encountered in the acute setting. They comprise a spectrum of diagnoses, including cellulitis, mastitis, and breast abscess. Although these conditions are typically seen in young women in the puerperium, postpartum, and lactation periods, they can also be nonpuerperal or have noninfectious causes (44). Risk factors for development of nonpuerperal breast infections are smoking, diabetes, obesity, and nipple piercing (45–48).

Hospital Logo

Patient Name _____ ID: _____ Gender: M/F Date of birth _____ Age _____

Date _____ Tech _____ Referring physician _____

Breast US Worksheet

CLINICAL INFORMATION (encircle all that apply):

Reason for exam: Breast pain (*focal, multifocal, diffuse, cyclical*) Breast lump Trauma Nipple discharge
(*white, yellow, green, clear, bloodstained; spontaneous, nonspontaneous; single orifice, multiple orifices*)
Other: _____

Side: Left Right Bilateral

History of breast surgery: No Yes If yes, please specify type and year: _____

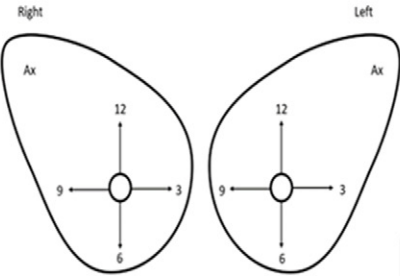
History of breast cancer: No Yes If yes, please specify year and treatment: _____

Family history of breast cancer: No Yes
If yes, please specify: mother, sister, grandmother, daughter, aunt, and age _____

Physical exam of the area of concern: Negative Palpable mass Palpable nodularity Skin thickening
Skin retraction Nipple discharge Nipple Retraction Scar
Other: _____

SONOGRAPHIC FINDINGS:

Right

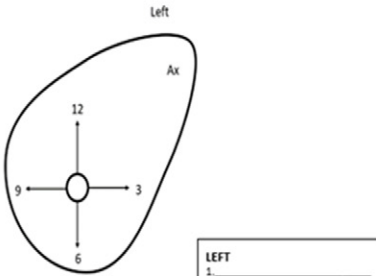


RIGHT

1. _____
Location (o'clock position, distance from nipple in cm): _____
Dimensions (cm): _____

2. _____
Location (o'clock position, distance from nipple in cm): _____
Dimensions (cm): _____

Left



LEFT

1. _____
Location (o'clock position, distance from nipple in cm): _____
Dimensions (cm): _____

2. _____
Location (o'clock position, distance from nipple in cm): _____
Dimensions (cm): _____

SONOGRAPHER'S IMPRESSION:

Figure 7. Proposed template for an ED US technologist form. Ax = axilla.

Cellulitis and Mastitis.—Cellulitis is an inflammatory-infectious process of the skin that may extend into the dermis and subcutaneous fat (Fig 9). With mastitis, the inflammatory changes extend to the breast parenchyma (Fig 10). US findings of cellulitis and mastitis include skin thickening, increased parenchymal echogenicity, edema, and increased vascularity on Doppler images (44,45,49). Owing to the significant overlap in the clinical and imaging findings of mastitis and inflammatory breast cancer (IBC), the most lethal form of breast cancer, close monitoring of symptomatic response to antibiotic therapy in patients with mastitis is strongly recommended (46,50).

Inflammatory Breast Cancer.—IBC is a highly aggressive invasive cancer that is associated with distant metastases in up to 40% of patients at presentation (50). Common clinical features include rapid onset of diffuse asymmetric breast enlargement, skin changes (eg, erythema with or without bruising, warmth, skin thickening resembling an orange peel or peau d'orange), and nipple abnormalities (eg, flattening or inversion) (8,50–52). Typical sonographic features of IBC are skin thickening, distended lymphatic vessels within the dermis, suspicious lymphadenopathy, heterogeneous breast parenchyma with or without solid masses, with diffuse edema, increased vascular-

ity, and lack of focal fluid collections (8,50–52). However, no clinical or sonographic pathognomonic feature of IBC exists. For this reason, in addition to clinical monitoring, radiologists should consider recommending a complete evaluation in the breast imaging department to document resolution of symptoms in patients who have a mastitis-like presentation and are due for screening mammography. Prompt complete evaluation of patients for whom there is high suspicion for IBC—namely, women who are of an older age (average age at IBC diagnosis, 58 years); are nonlactating; have a family or personal history of breast cancer; and/or have any or a combination of regional adenopathy, peau d'orange, nipple flattening or inversion, and diffuse breast parenchyma involvement of at least one-third of the breast at initial presentation—should also be recommended (8,45,46,50,51). Timely complete breast evaluation, often with MRI and breast biopsy, is also indicated for patients with a poor response or no response to antibiotic treatment (46,51).

Breast Abscess.—Breast abscesses, localized purulent collections that typically occur as complications of mastitis, usually manifest as heterogeneous fluid collections that have internal echoes and thickened irregular walls, without internal vascularity, on US images (Fig 11) (44,45). When a fluid collection

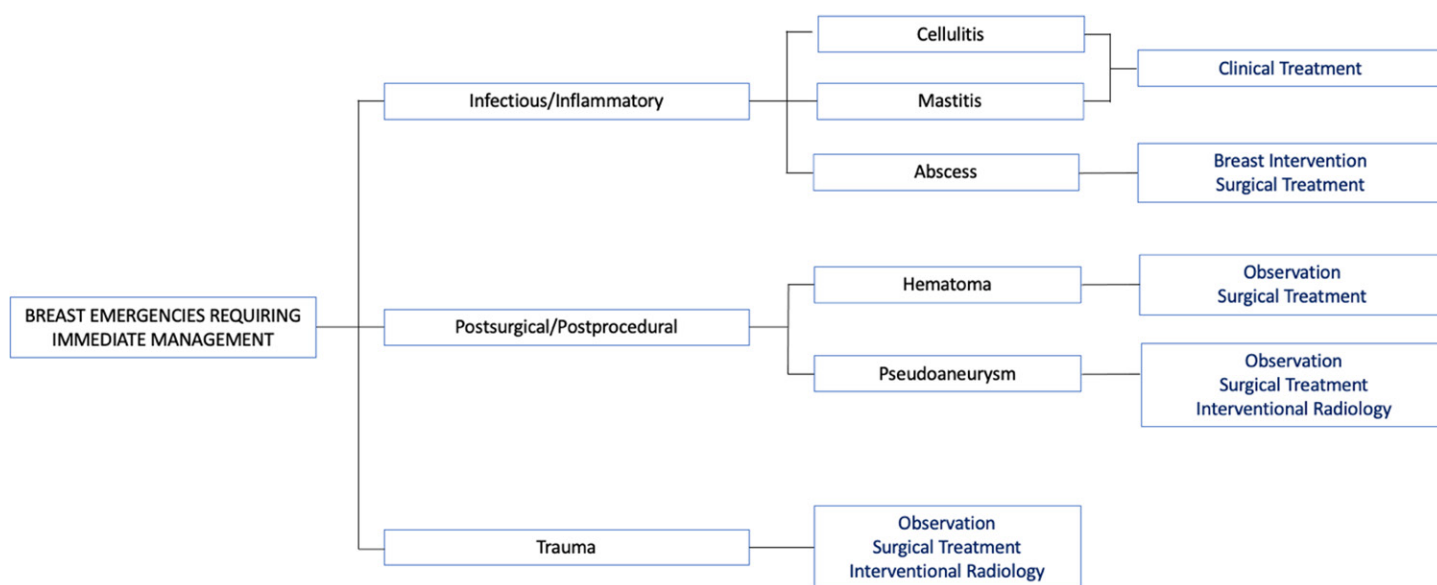


Figure 8. Common breast conditions in the acute care setting that require immediate management.

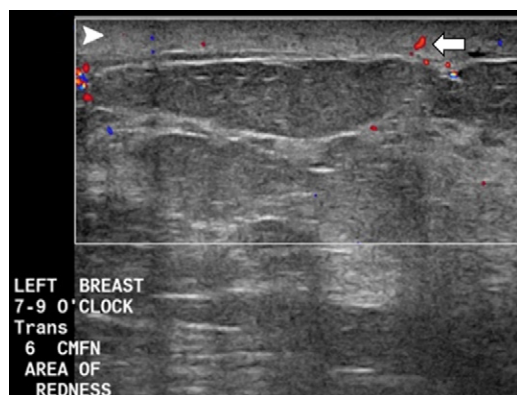


Figure 9. Palpable area of thickening, erythema, and swelling of the left breast for the past 3 days in a 33-year-old woman in the 18th week of pregnancy who presented to the ED. Color Doppler US image shows mild skin thickening in the area of concern (arrowhead), with mild heterogeneity and minimal increased vascularity (arrow). Note the lack of subcutaneous edema or fluid collections, which differentiates this mass from mastitis or abscess.

is identified in the breast of a patient with clinical symptoms of infection, an abscess should be suspected and urgent intervention with drainage (imaging guided or surgical) should be recommended. At US, abscesses may have posterior acoustic enhancement, a thick irregular echogenic rim, and increased peripheral vascularity, hampering their differentiation from necrotic or cystic tumors (Fig 12) (44,45). Studies (8,53) have demonstrated an association of posterior acoustic enhancement with high-grade and hormone receptor–negative breast tumors, causing additional overlapping of imaging features between these neoplasms and breast abscesses.

Management of breast abscesses involves a combination of antibiotics and drainage or aspiration. Clinical follow-up may be sufficient for lactational abscesses with a complete response to initial treatment (45). After completing antibiotic

therapy, patients with nonpuerperal abscesses should be referred to the breast imaging department for clinical and US follow-up to ensure complete resolution of symptoms (8,45). Mammography is recommended for patients older than 30 years and cases of nonpuerperal abscesses and should be performed after the acute phase has resolved (8,44,45,48). Biopsy is indicated if a poor response or no response to treatment is observed (45,48).

Other Inflammatory Conditions.—Other benign breast conditions that may mimic infectious cellulitis, mastitis, and/or IBC clinically and sonographically include lymphocytic (diabetic) mastopathy; postradiation changes; vascular abnormalities, including venous hypertension or lymphatic obstruction; sarcoidosis; amyloidosis; connective tissue diseases (eg, systemic lupus erythematosus, Churg-Strauss syndrome, granulomatosis with polyangiitis); and idiopathic granulomatous mastitis, which can also mimic breast abscesses. In these cases, tissue biopsy may be required to rule out malignancy and make a definitive diagnosis (Table 2) (8,44,46,48,52).

Mondor Disease.—Mondor disease, or superficial thrombophlebitis of the thoracoepigastric vein, is a rare cause of acute breast pain (49,54). At physical examination, it manifests as a palpable tender erythematous cordlike subcutaneous structure extending from the anterior inferior aspect of the breast to the epigastrium. US is the modality of choice for detection of this entity and typically shows an anechoic tubular structure with a beaded appearance. Intraluminal clots and associated mild surrounding soft-tissue inflammatory changes may be present. Mondor disease is a self-limiting condition, and imaging follow-up is not required (49,54).

Postsurgical and Traumatic Injuries

The most frequent complications encountered during the early postoperative period after breast surgery are fluid collections, such as seroma and hematoma, and infection (54,55).

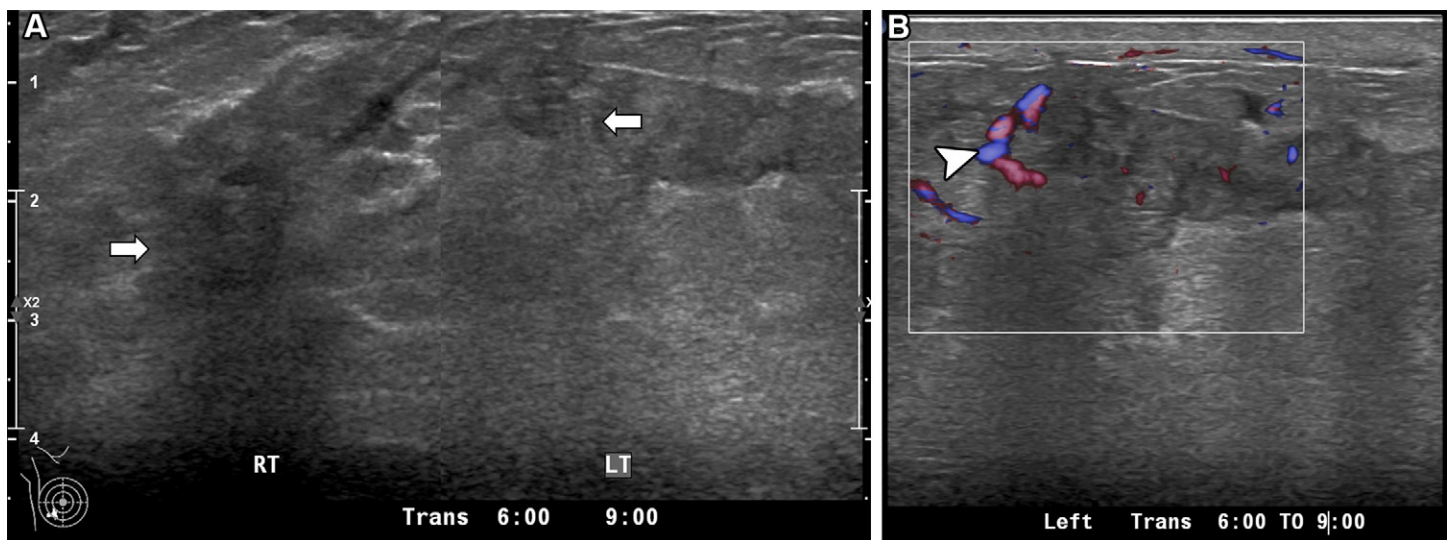


Figure 10. Bilateral breast pain with fever and erythema in a 25-year-old woman who was lactating and presented to the ED. Initial transverse gray-scale (A) and color Doppler (B) US images obtained in the ED show areas of hypoechoogenicity (arrows in A) and increased vascularity (arrowhead in B) in both breasts. No fluid collections or suspicious masses were identified. The findings were suggestive of mastitis, and the patient was treated with antibiotics and followed up clinically until the symptoms resolved. Clinical follow-up must be performed until symptom resolution.

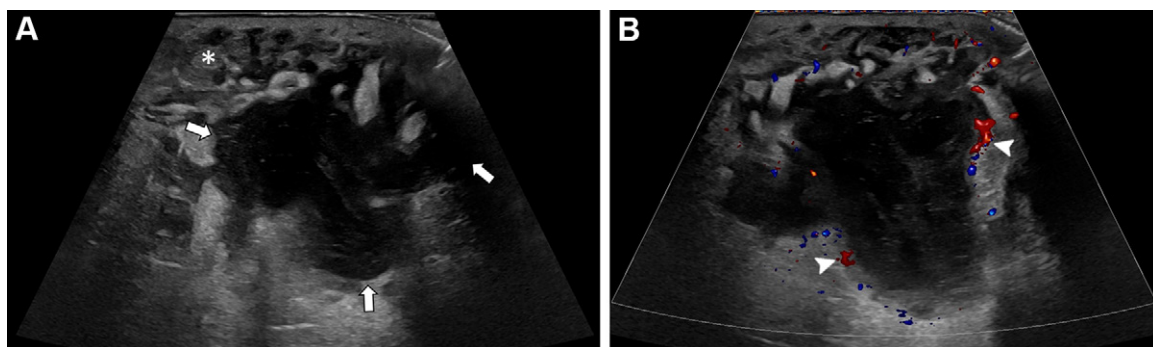


Figure 11. Palpable abnormality in the right breast, with associated pain and fever, in a 34-year-old woman who presented to the ED. No additional clinical information was provided. Initial gray-scale (A) and color Doppler (B) US images obtained in the ED show an irregular hypoechoic to anechoic collection (arrows in A) in the right breast with surrounding vascularity (arrowheads in B). No suspicious solid components or internal vascularity were visualized. Note the significant associated edema (*) in A) in the subcutaneous soft tissues. This patient was diagnosed with a breast abscess, which was treated emergently by the surgical team with incision, drainage, and antibiotics. When a breast abscess is encountered, imaging-guided intervention is an alternative treatment option. Affected patients require clinical follow-up until symptoms resolve, as well as evaluation by breast imaging department staff after antibiotic treatment is completed.

Seroma.—Seromas are serous fluid collections in the post-surgical cavity that most frequently occur after lumpectomy, mastectomy, and/or axillary surgery (55). They can be seen in up to 50% of modified mastectomy cases (55). At US, the appearance of seromas can vary from anechoic to heterogeneous fluid collections (Fig 13). Although they are usually self-limited, some cases of seroma may persist for years and cause symptoms such as a palpable abnormality and pain. Percutaneous aspiration may be considered, depending on the size and severity of the patient's symptoms, and patients may be followed up clinically by their managing physician and with imaging according to their regular screening schedule (54–56). If clinically warranted, aspiration may be performed to collect specimens for culture or to allow faster healing (55,56). Rarely, seromas can become infected or evolve into hematomas after percutaneous aspiration or direct trauma (Fig 14) (56).

Hematoma.—Breast hematomas are collections composed of blood products that are usually encountered following surgery, percutaneous biopsy, or acute trauma. They may be present in 2%–10% of breast surgery cases (55) and at 0.7%–23% of biopsies performed with stereotactic vacuum-assisted techniques (57).

Clinically, a breast hematoma may manifest as a new breast swelling or palpable lump that is expanding or stable, with or without overlying skin discoloration (8,44,49). The appearance of hematomas on US images depends on their stage; they may appear as hypoechoic fluid collections with subtle internal debris in the acute stage or as a complex heterogeneous mass with septa in the subacute stage (Fig 15) (8). Chronic organized hematomas usually appear as encapsulated masses with heterogeneous echogenicity. However, if a hematoma undergoes liquefaction, it can appear as anechoic or hypoechoic fluid collections in the chronic stage (58). Mild vascular flow

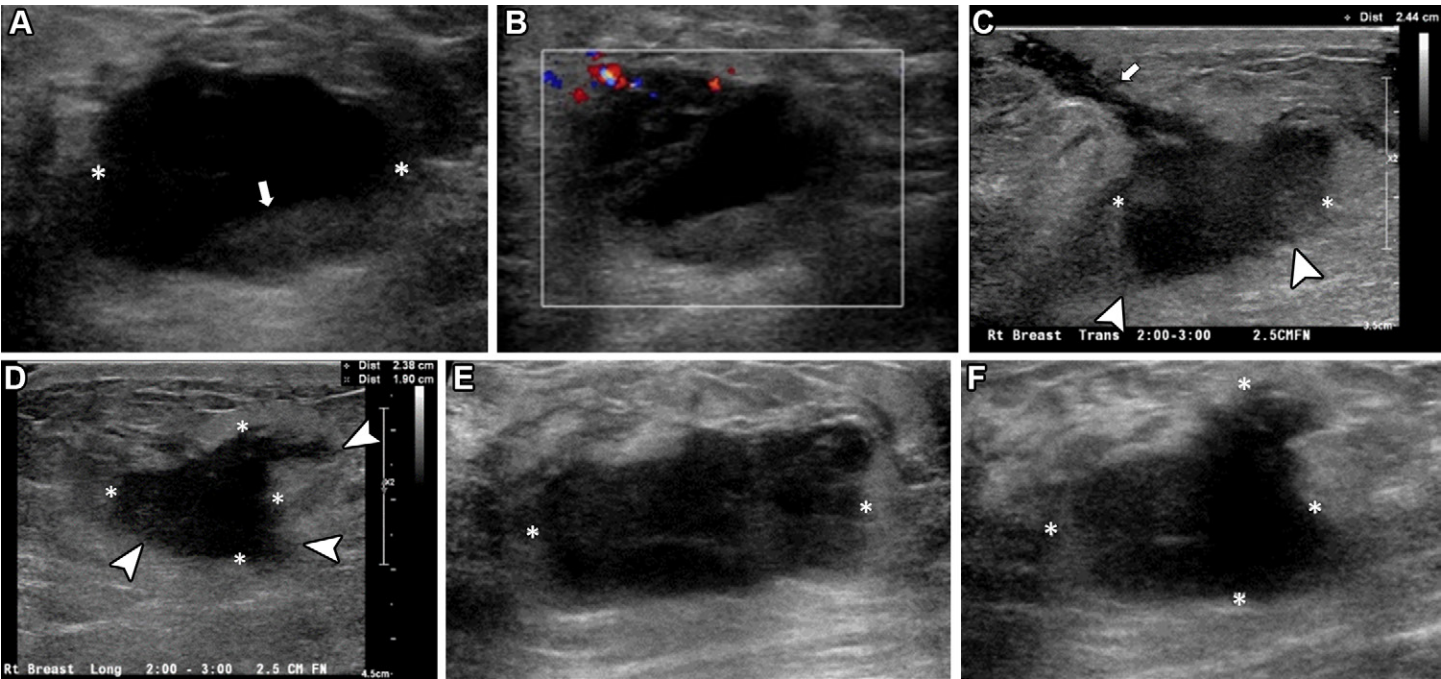


Figure 12. Palpable lump and pain in the right breast in a 37-year-old woman who was evaluated in the ED. (A, B) Initial gray-scale (A) and color Doppler (B) US images show an oval hypoechoic to anechoic lesion (*) in the right breast, which appeared to have a fluid-fluid level (arrow in A) and some peripheral vascularity detected (B). An abscess was suspected, and needle aspiration without imaging guidance was attempted in the ED. However, no obvious aspirate was obtained. The patient was managed clinically with antibiotics, but she returned to the ED 1 month later owing to persisting pain and new sero-sanguineous drainage through the skin. (C, D) Orthogonal US images obtained at the second visit had a better gain adjustment, showing the mass (*) with additional features such as an irregular shape and indistinct margins (arrowheads). A tract (arrow in C) toward the skin in the region of active drainage was observed. (E, F) US images obtained at recommended urgent complete assessment in the breast imaging department show a solid, irregular, hypoechoic mass (*) with angular and indistinct margins at the site of concern. Biopsy revealed triple-negative invasive ductal carcinoma.

Table 2: Clinical and US Features of Breast Conditions That Mimic Breast Cancer		
Breast Condition	Clinical Presentation	US Findings
Inflammatory and/or infectious		
Infectious mastitis	Skin erythema, skin thickening, skin warmth, breast swelling, lymphadenopathy	Increased vascularity, distended lymphatic channels, skin thickening
Abscess	Palpable mass, skin erythema, skin thickening, skin warmth, lymphadenopathy	Irregular shape, thick walls with increased vascularity, complex internal echo pattern, posterior acoustic enhancement
Lymphocytic mastopathy	Hard palpable mass, skin thickening	Irregular shape, hypoechoic mass, indistinct margins, posterior acoustic shadowing, increased vascularity
Granulomatous mastitis	Palpable mass, diffuse breast swelling, skin erythema, lymphadenopathy	Irregular shape, hypoechoic mass, clustered hypoechoic tubular tracks, increased vascularity
Posttraumatic and/or postsurgical		
Hematoma	Palpable mass	Irregular shape, thickened walls with nodularity, solid components, broad septa, internal vascularity in chronic organizing hematomas
Fat necrosis	Palpable mass, skin retraction	Irregular shape, posterior acoustic shadowing

may be detected in chronic organized hematomas; however, no internal vascularity should be present in the acute and sub-acute stages (55,58).

Most hematomas are self-limited. However, large or rapidly expanding hematomas may require surgical evacuation (44,49,55). Imaging findings of hematomas, such as mild internal vascularity during the chronic stage, irregular shape, thickened walls with nodularity, solid components, and broad septa, may mimic breast cancer (59). Clinical observation and imaging follow-up in the breast imaging department should be performed to document resolution, especially in patients who have spontaneous and persistent hematomas (ie, no clinical resolution within 4–6 weeks) and those who are due for screening mammography (54).

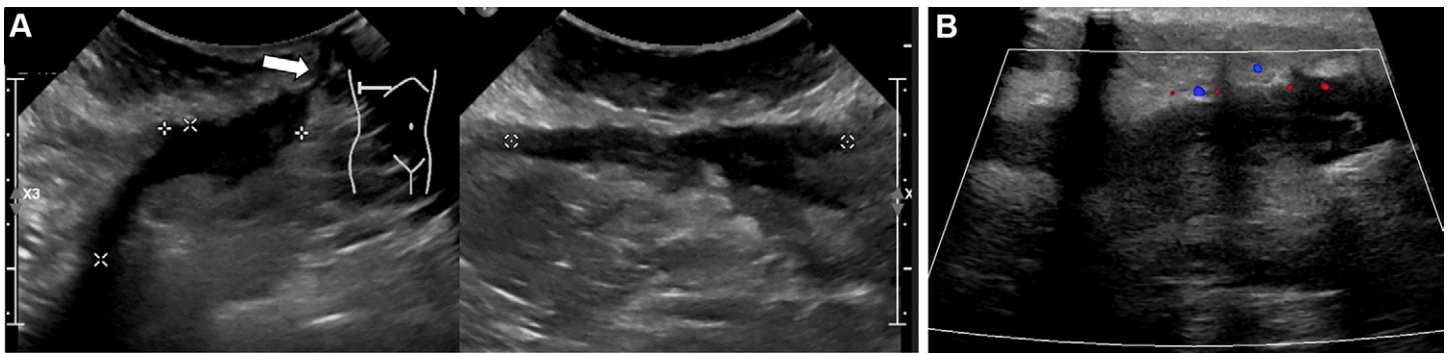


Figure 13. Active drainage of the right chest wall in a 58-year-old woman after right mastectomy who presented to the ED. (A) Initial transverse (left) and longitudinal (right) US images for evaluation in the ED show a hypoechoic fluid collection (calipers) in the surgical bed, tracking to the skin (arrow). (B) On a color Doppler US image, no suspicious masses or increased vascularity in the surrounding soft tissues was identified to suggest superimposed infection. Subsequent drainage in the breast imaging department revealed a seroma due to recent mastectomy. Observation is an alternative for patients with postsurgical seromas, and expectant treatment should be considered according to the size and severity of the patient's symptoms.

Pseudoaneurysms.—Pseudoaneurysms in the breast are rare and predominantly associated with a history of biopsy, surgery, or trauma. They are usually associated with hematomas, and a palpable pulsatile mass is often identified at physical examination (44). US evaluation is usually sufficient for the diagnosis, demonstrating a well-defined anechoic structure with a narrow connection to the adjacent artery and a characteristic bidirectional swirling blood flow pattern at color Doppler imaging (44,60). Treatment options include compression dressing and observation, US-guided manual compression, thrombin or alcohol injection, coil embolization, and surgical repair (Fig 16) (44,60).

Implant Rupture.—Breast augmentation is the most common cosmetic surgical procedure in the United States, with a 45% increase in the prevalence of this surgery over a decade from 2008 (61). A common implant-related complication that may be encountered in the ED is implant rupture, which can be related to damage caused by one or more surgical instruments during implant placement, trauma, or implant failure, with the latter usually associated with implant aging. Although most ruptures are clinically silent, acute symptoms such as breast erythema, swelling, and/or hardening; change in breast shape; and painful axillary lymph nodes may be present (62).

Ruptures of saline implants are clinically evident; however, silicone implant ruptures have subtle clinical findings and may require imaging. Echogenic lines visualized within the implant lumen that may be contiguous or discontinuous with the implant envelope are characteristic of the “stepladder” sign and suggest an intracapsular silicone implant rupture (Fig 17) (62,63). The “snowstorm” sign, with which free silicone droplets in the breast parenchyma or axillary lymph nodes are depicted as hyperechoic shadowing structures on US images, suggests an extracapsular implant rupture (62,63). Patients require treatment for their symptoms, with reassurance that neither saline leakage nor silicone leakage poses imminent or significant health risks, and must be referred to the breast department for further evaluation. Treatment involves observation, implant removal, or implant replacement (62,63).

Foreign Bodies.—Common reasons for ED visits include penetrating injuries and suspected retained foreign bodies (59). Surgical and biopsy clips and fragments of hook wires are the most common foreign bodies seen in the breast (64). Pieces of glass, bullets, needles, and gauze have been also reported (59,64). US enables the detection of different radiolucent foreign bodies and aids in localization of radiolucent and radiopaque foreign bodies and evaluation of associated soft-tissue complications (59). In acute settings, foreign bodies usually appear on US images as echogenic structures with posterior acoustic shadowing or reverberation artifact and a hypoechoic halo secondary to a surrounding inflammatory reaction (59). Long-term sequelae such as foreign-body granulomatous reaction and fat necrosis may appear as a palpable mass with skin dimpling or retraction and suspicious US features, raising concern for neoplasm, especially when the history of the previous trauma or surgery is lacking and a foreign body is not recognized at imaging (49,64).

Nonurgent Breast Conditions in the ED

Reports of breast pain, palpable masses, nipple discharge non-related to trauma, and findings suggestive of postoperative complications and infection also can be encountered in the ED (9,13,49). While these entities are worrisome for the patient, they may not require immediate intervention in the ED. However, the radiologist must consider them in the differential diagnoses to avoid delays in appropriate management and follow-up (Fig 18).

Breast Pain

Breast pain is a common clinical symptom that affects up to 70% of women during their lifetime (65). Although the probability of breast cancer in patients with breast pain as the only symptom is low, imaging evaluation may reassure these patients and help diagnose potentially treatable abnormalities (6). Patients who present with diffuse, bilateral, cyclical pain can be safely referred for outpatient evaluation, as this feature is benign and related to hormonal changes (6).

Breast Cysts

Simple cysts are round well-circumscribed anechoic structures with posterior enhancement and thin walls and demonstrate

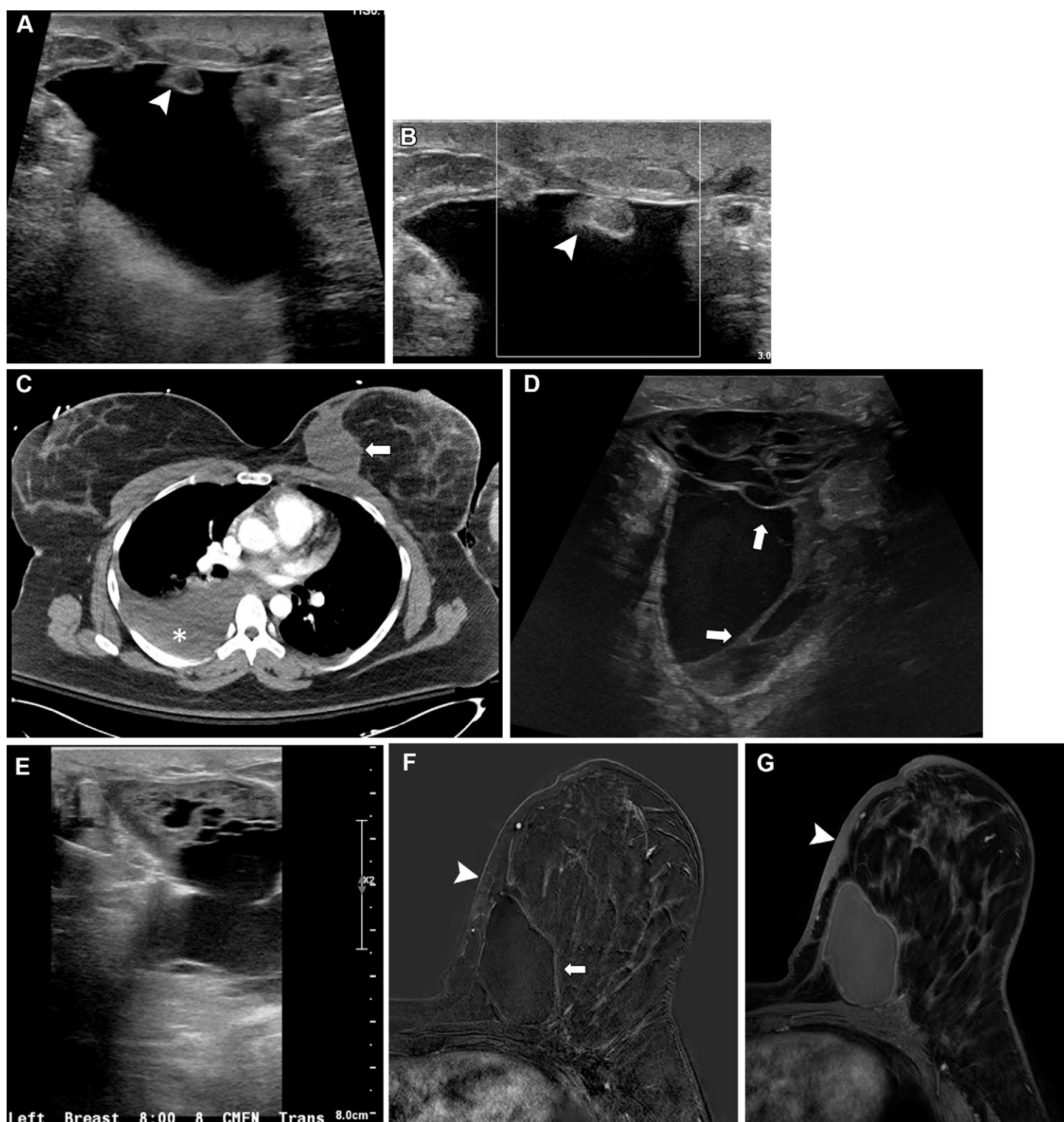


Figure 14. Chest and back pain in a 34-year-old woman who presented to the ED after a motor vehicle accident. She had a history of left breast cancer treated with left lumpectomy several years earlier. (A, B) Recent US images prior to presentation show a seroma, which contains mobile debris (arrowhead) and has no internal vascularity (B), in the lumpectomy bed. (C) Axial CT image obtained as part of the trauma workup shows a high-attenuation fluid collection (arrow) in the medial aspect of the left breast and a posttraumatic right hemothorax (*). At discharge, the patient was referred for follow-up in the breast imaging department. At follow-up 6 weeks later, she reported having a persistent lump and pain in her left breast. (D) Follow-up US image shows a large complex fluid collection with thick septa (arrows), internal debris, and no obvious increased vascularity, consistent with a seroma, which was probably complicated by superimposed subacute hematoma formation. (E) US-guided aspiration was performed for symptom relief, and cytologic analysis results were negative for malignancy. (F, G) Subtraction (F) and postcontrast T1-weighted fat-saturated (G) MR images obtained 4 months later as part of the patient's surveillance show a benign seroma (arrow in F) without suspicious enhancement on the subtraction image and skin thickening (arrowhead) secondary to radiation changes affecting the left breast on the postcontrast T1-weighted fat-saturated image.

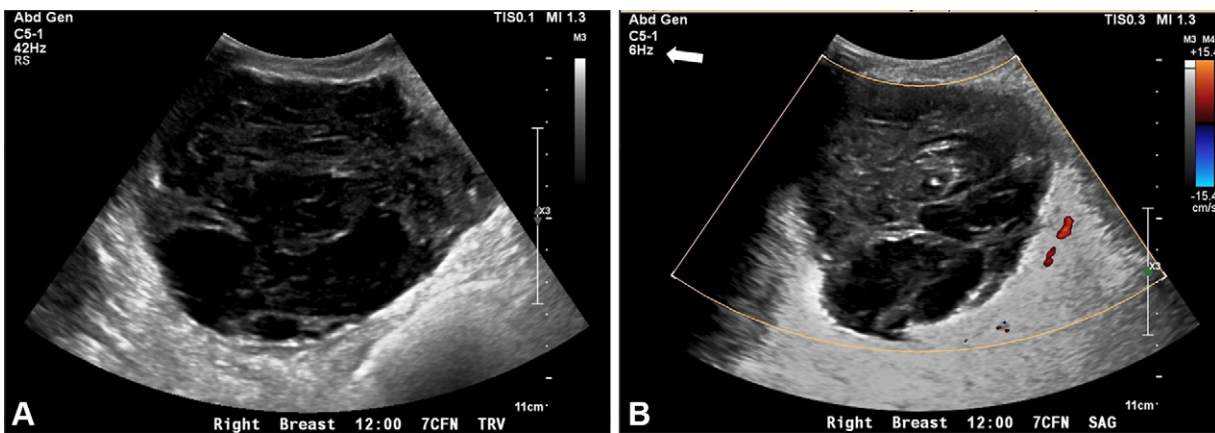


Figure 15. Palpable abnormality in the right breast after a fall in a 63-year-old woman who presented to the ED. (A) Initial US image shows a heterogeneous, complex, solid-cystic mass in the right breast. (B) Color Doppler US image shows no internal vascularity in the solid components. Despite the initial suspicious appearance of the mass, a hematoma was considered, given the patient's clinical history and presentation. She was observed clinically in the ambulatory setting until her findings resolved. In this case, a 6-Hz (arrow) transducer was used to evaluate the abnormality owing to its size.

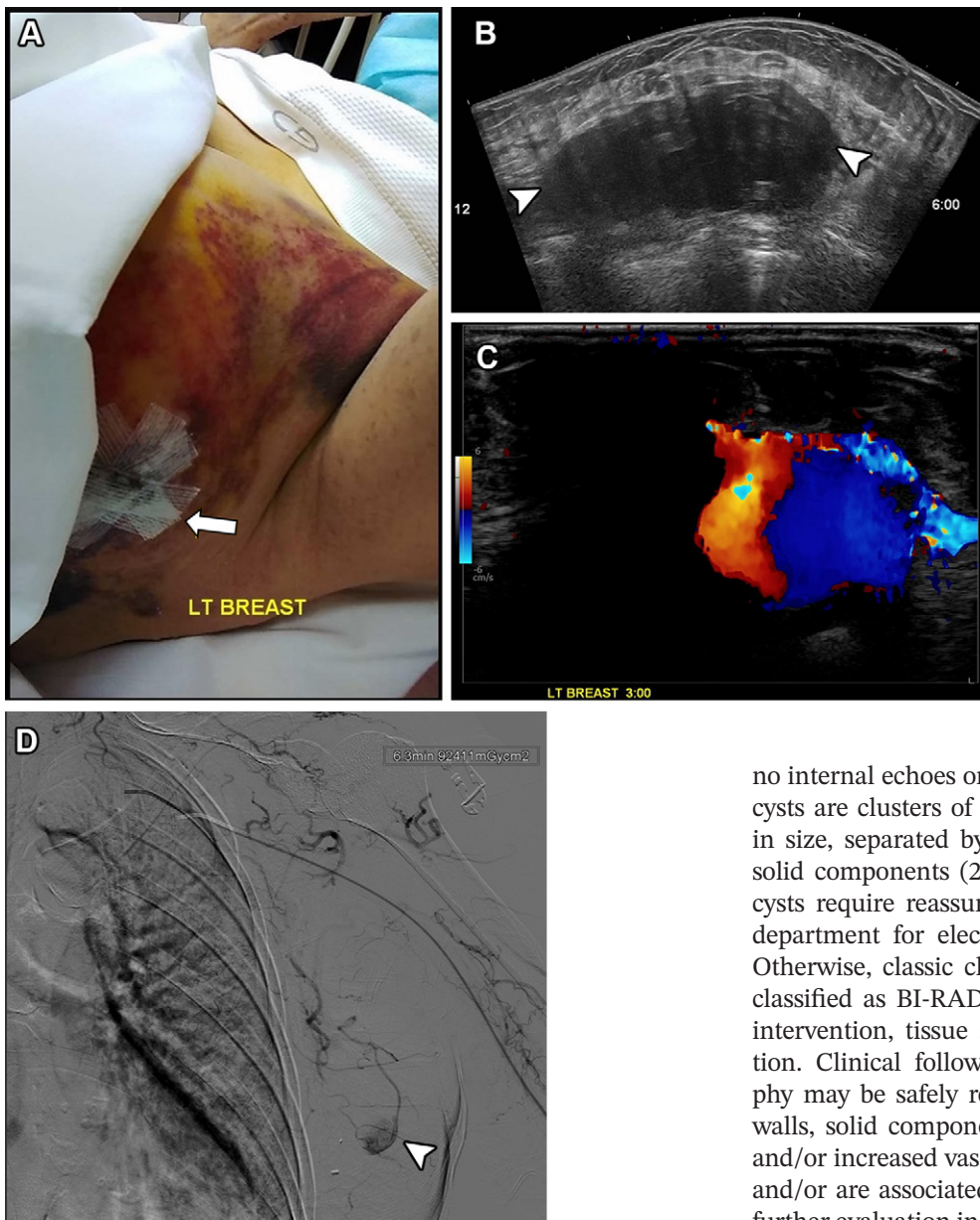


Figure 16. Left breast swelling and pain after recent stereotactic biopsy in a 68-year-old woman who presented to the ED. (A) Photograph shows physical examination findings described by the technician, which included a large bruise on the left breast associated with the biopsy entry site (arrow). (B) Initial US image obtained in the ED shows an oval mixed hypoechoic to anechoic fluid collection (arrowheads) in the left breast. Hematoma was diagnosed on the basis of the patient's clinical history and clinical findings. (C) Further evaluation with color Doppler US revealed an associated highly vascularized structure with bidirectional flow, suggestive of a pseudoaneurysm. (D) Emergent embolization of a distal segment of the lateral thoracic artery (arrowhead) was performed in the interventional radiology department.

no internal echoes or solid components (25). Clustered microcysts are clusters of anechoic foci, each of which is 2–3 mm in size, separated by thin (<0.5-mm) septa without discrete solid components (28,37). Patients with large painful simple cysts require reassurance and may be referred to the breast department for elective aspiration for symptom relief (66). Otherwise, classic clustered microcysts and simple cysts are classified as BI-RADS category 2 lesions and do not require intervention, tissue sampling, or additional imaging evaluation. Clinical follow-up and routine screening mammography may be safely recommended (66). Cysts that have thick walls, solid components, an irregular shape, internal debris, and/or increased vascularity in the wall or surrounding tissue, and/or are associated with axillary lymphadenopathy require further evaluation in the breast imaging department (66).

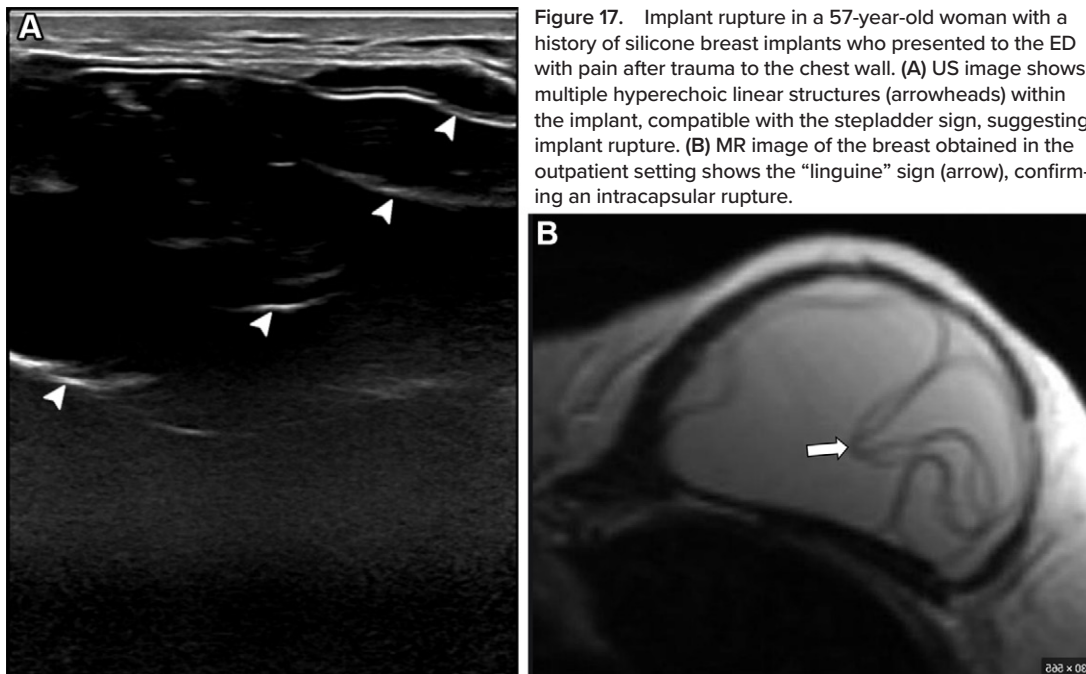


Figure 17. Implant rupture in a 57-year-old woman with a history of silicone breast implants who presented to the ED with pain after trauma to the chest wall. (A) US image shows multiple hyperechoic linear structures (arrowheads) within the implant, compatible with the stepladder sign, suggesting implant rupture. (B) MR image of the breast obtained in the outpatient setting shows the “linguine” sign (arrow), confirming an intracapsular rupture.

Palpable Masses

Palpable abnormalities require an initial clinical breast examination by the ordering provider to specify pertinent findings in the study request. The clinical features of masses can aid in differentiating benign versus malignant masses, but these features alone cannot exclude malignancy, and, thus, imaging is indispensable. Malignant masses may be hard, irregular in shape, fixed to adjacent tissues, and associated with skin thickening, dimpling, or retraction (7,67). On the other hand, benign masses are typically soft and mobile, with well-defined margins (7,67). The sonographer must report these clinical observations by documenting them on the technologist form (10).

Probably Benign Masses.—Palpable abnormalities with benign clinical features and US findings consistent with simple cysts, normal intramammary lymph nodes, or typical sebaceous cysts may be categorized as benign, and routine mammographic screening may be recommended (5,7,67).

When solid masses are encountered, features that suggest benignity include oval shape, circumscribed margins, orientation parallel to the skin surface, no acoustic shadowing, and homogeneous isoechoic or hypoechoic internal echotexture (Fig 19) (25). These characteristics must be concordant with benign clinical features at the physical examination. If a probably benign mass is encountered in the ED, the patient must be referred to the breast imaging department for complete evaluation and BI-RADS characterization, regardless of his or her age. Patients who are aged 30 years or older require additional workup with mammography, while those younger than 30 years require baseline US evaluation to establish the proper imaging follow-up (25).

Suspicious Masses.—An initial diagnosis of breast cancer in the ED is uncommon, occurring in fewer than 10% of all breast cancer cases (68). However, it is crucial to recognize

this malignancy in the ED, as it is associated with poorer outcomes compared with breast cancer diagnosed in the outpatient setting and tends to affect women with sociodemographic inequalities disparately (69).

US features that raise concern for malignancy include irregular shape; antiparallel orientation; spiculated, microlobulated, angular, or indistinct margins; heterogeneous echotexture; and posterior shadowing (Fig 20) (25). If any of these imaging characteristics is present, the abnormality must be categorized as suspicious. In addition, suspicious imaging findings take precedence over clinical findings in cases in which the latter are not concerning, and urgent evaluation in the breast imaging department for complete imaging assessment and possible biopsy is necessary for a patient of any age. Involving the navigation team and scheduling the follow-up appointment before the patient is discharged from the ED can aid in timely evaluation and continuity of care (29,33).

Negative Findings

Patients may present to the ED with diverse breast symptoms and no imaging correlation. These cases can be approached according to the patient's clinical presentation and age. Patients of any age who have clinical findings that are concerning for malignancy but negative US findings must be promptly referred to the breast imaging department for complete imaging evaluation and possible biopsy. This includes patients with nipple discharge that demonstrates at least one of the following suspicious features: bloody or clear, arising from a single orifice, initially manifesting as spontaneous, and unilateral (70). These patients are often found to have negative imaging findings at presentation and require prompt complete imaging workup.

Symptomatic patients who are aged 30 years or older and have negative US findings are considered to have had an incomplete assessment and require proper evaluation in the

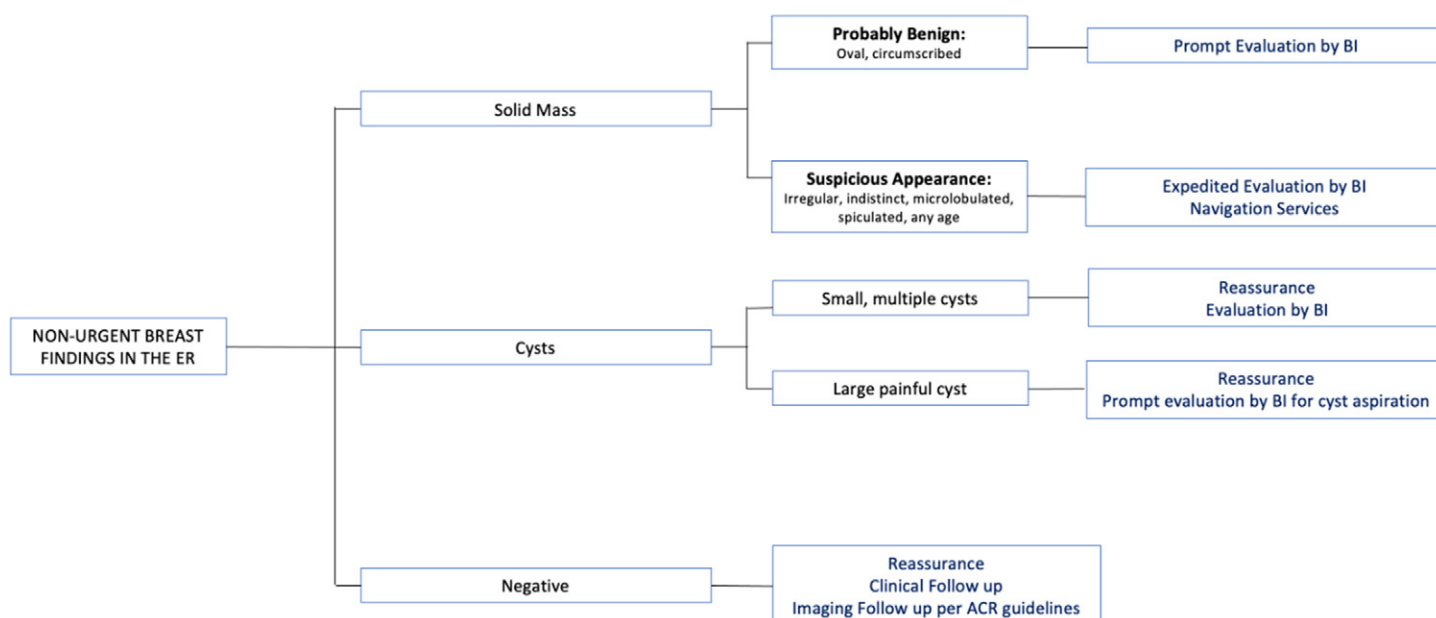


Figure 18. Common imaging findings seen in the acute care setting for which immediate management is not imminent but complete assessment in the breast imaging department (BI) may be required. Note that for suspicious abnormalities, expedited evaluation is necessary. ACR = American College of Radiology, ER = emergency department.

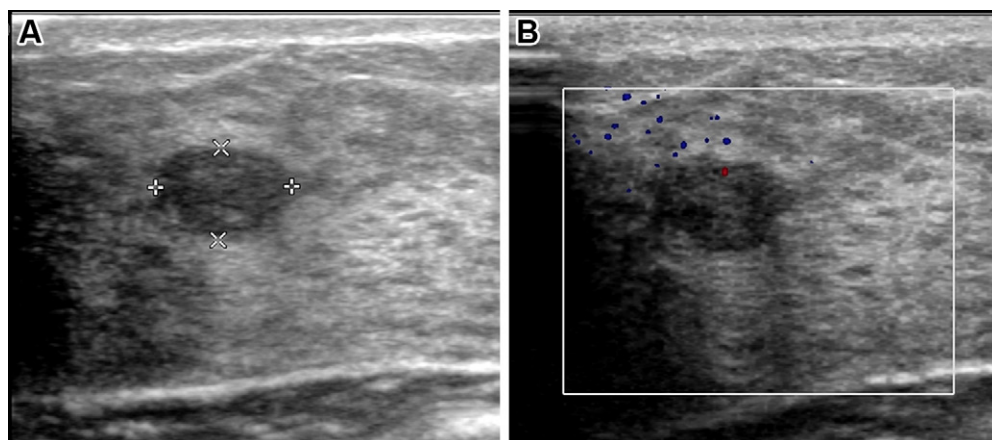


Figure 19. Right breast lump, first noticed earlier in the week, in a 20-year-old woman, which prompted her to seek consultation in the ED. Initial gray-scale (A) and color Doppler (B) US evaluation in the ED revealed an oval hypoechoic mass (calipers in A) in the right breast, without suspicious features or increased vascularity (B). On the basis of the US features, the mass was believed to be a fibroadenoma. Complete evaluation with US in the breast imaging department was recommended. Benign masses and cysts can be encountered in the ED, and appropriate follow-up must be recommended in such cases.

breast imaging department (5,7,70). Women with an average risk of cancer who are younger than 30 years and have negative US findings and nonsuspicious clinical findings may be reassured that the findings are not suggestive of breast cancer and be followed up clinically by their clinician (5–7,70). Women who are younger than 30 years and have a high risk for breast cancer, including those who have a personal history or strong family history of breast cancer, have undergone prior mantle-field radiation therapy, and have genetic mutations that predispose them to breast cancer, should be referred to the breast imaging department (4).

Conclusion

Imaging evaluation of patients who present to the ED with breast symptoms is restricted by the absence of mammographic units, the lack of technologists who are experienced and in proximity to the radiologist, and inadequate access to patients' complete clinical and medical histories. These limitations may result in inaccurate, incomplete, and

delayed diagnoses. Efforts to overcome these challenges, such as improving radiologist-technologist communication, technologist training, availability of necessary equipment, and systems for expedited interpretation and patient follow-up, result in a better quality of care for these patients. The goals for radiologists in the ED are to understand these challenges, actively participate in strategies to overcome them, and more importantly, appropriately identify those cases that require urgent intervention so that accurate recommendations can be made and potential complications and delayed breast cancer diagnoses can be avoided.

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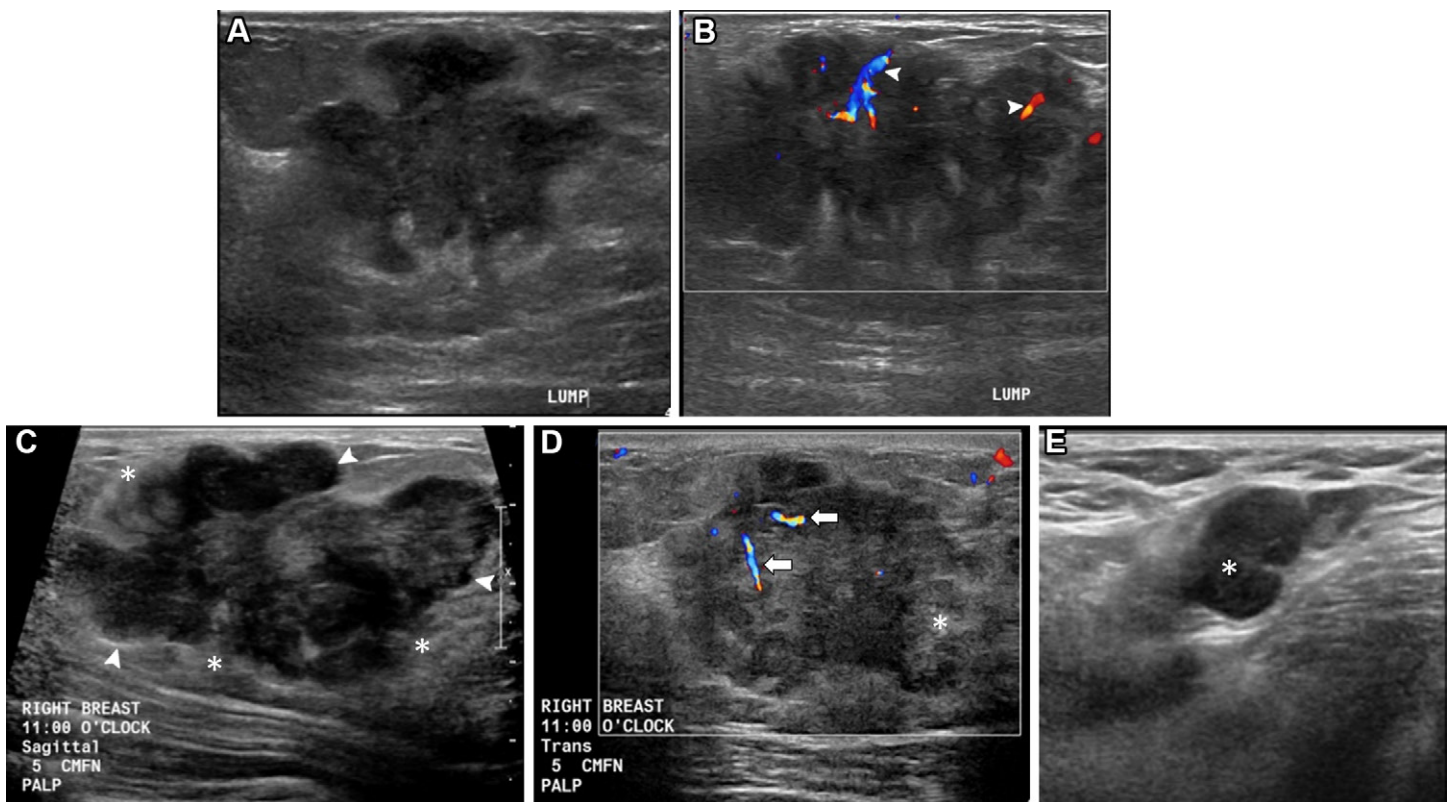


Figure 20. Pain and inflammation of the right breast for 4 weeks in a 30-year-old woman who presented to the ED. She had been lactating for the past 2 years. (A) Gray-scale US image obtained in the ED shows an irregularly shaped hypoechoic mass with an angular margin in the area of the palpable lump. (B) Color Doppler US image shows increased vascularity (arrowheads) within the mass. Expedited assessment in the breast imaging department was recommended. (C–E) US images obtained in the breast center confirmed the presence of the irregular mass, which had microlobulated (arrowheads in C) and indistinct (* in C and D) margins, internal vascularity (arrows in D), and an enlarged right axillary node (* in E). Biopsy results revealed grade II invasive ductal carcinoma, with a metastatic right axillary lymph node. Young patients who are not within the recommended age range for routine mammographic screening evaluation often have consultation in the ED for breast-related findings. Breast cancer can occur in young and lactating women, despite other diseases such as infectious and inflammatory entities being more common in these patient groups.

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